

Tuning Audio Control Server (ACS)

Terminology

- ACS - Tuning Audio Control Server.
- DAP - Dirac Audio Processor.
- Dirac - IO: A protocol specified by Dirac. Uses GRPC.
- GRPC - An open source Remote Procedure Call framework.
- Flow - Diracs audio configuration application.
- FilterLab - Diracs filter design application.
- DAPCTL - DAP Control.
- DAR - Dirac Archive, a format used to store the complete tuning configuration.

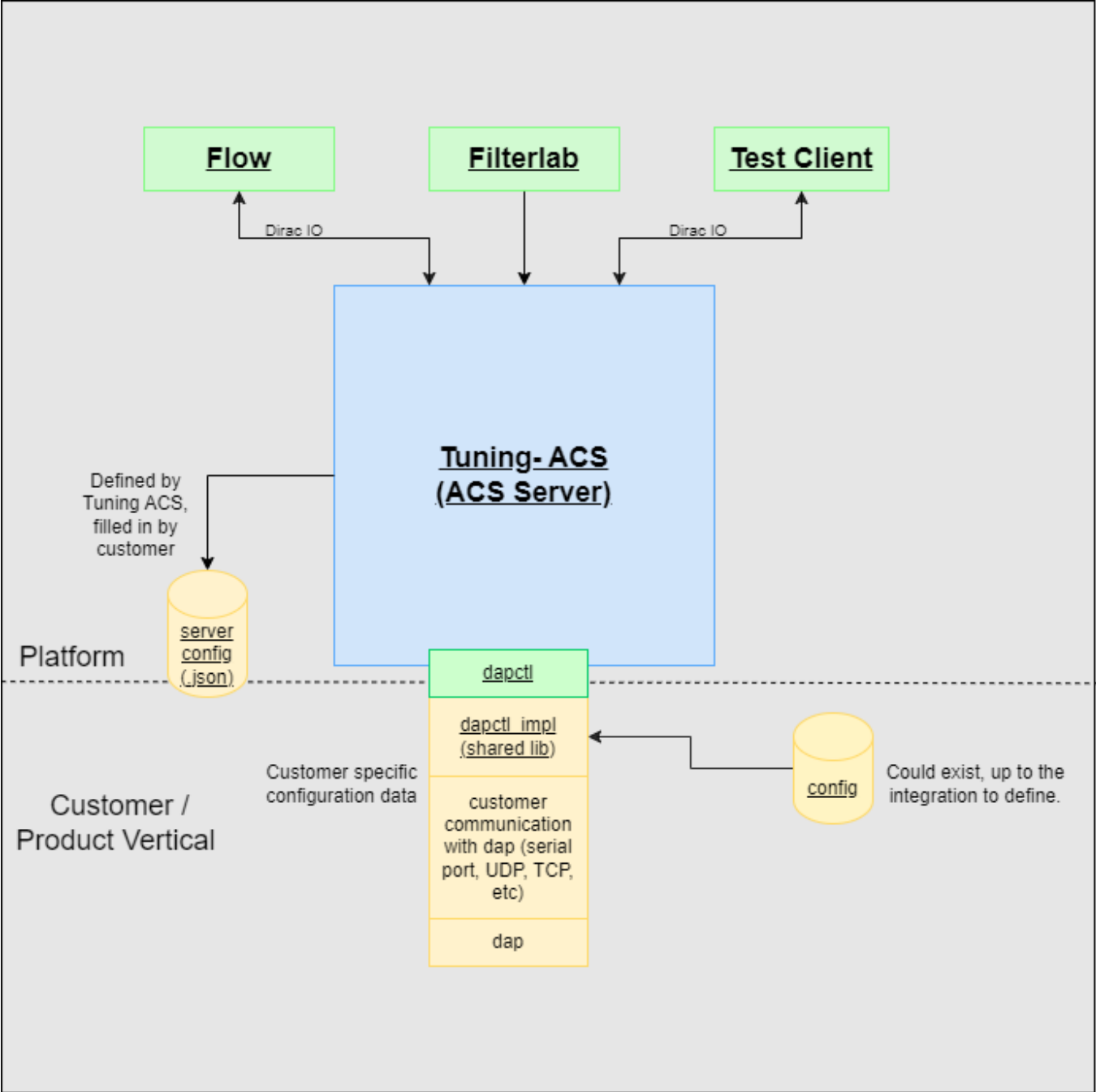
Background

The responsibility of the ACS is to facilitate the communication between the audio configuration application, Flow, and the on target integration, the DAP.

The type of communication includes:

- DAR module parameter control for real-time tuning,
- DAR file transfer,
- probe data transfer,
- measurements with FilterLab.

The picture belows illustrates a complete IAP integration with the ACS in the center.



The ACS itself acts as a server for Dirac IO. The ACS also acts as a client to the DAP.

The ACS can run on the same computer as Flow or it run on the target device. This will depend on the integration.

Configuration File (config.json)

This file is a component of tuning-acss package which provides all the necessary data needed by Flow/Filterlab.

It provides details on socket connection specifications, supported tuning and measurement capabilities, and device information.

Presently, the tuning-acss package comprises of an example-config.json file, which may differ across platforms.

Refer to the description below for the parameters of the example-config.json file.

Parameters	Description
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Parameters	Description
dap_uri	Pass the information to hardware, read by dapctl(Optional)
default_configuration	Sets the default DAR file configuration
devinfo	Holds details about the device
grpc_port	Port where the ACS server is listening
upnp_enabled	Indicates whether the UPnP protocol is enabled
use_normalized_parameters	Specifies the scale for tuning parameters
network_interface_name	Identifies the logical name of the network interface in use

DAPCTL

DAPCTL, short for DAP Control, serves as the runtime API whose primary function is to implement the communication between the ACS server and the DAP.

Operating in the ACS server backend, DAPCTL could either be running on a host or target environment, depending on the platform's framework.

It uses the platform's framework to facilitate communication between DAP and ACS server, thereby making it dependent on the platform.

ACS usage

The acs server is a command line application and its usage is explained in the [ACS server](#) user guide.

There are currently 3 different clients used to communicate with the acs server over dirac-io:

- [Test client](#) Command line test tool
- [Flow](#) Used for realtime tuning
- [Filterlab](#) Used for measurement

User the links above for a user guide regarding the different client applications.